Section A

Answer **all** the questions in this section.

1 Fig. 1.1 is an electron micrograph of part of a eukaryotic cell with organelles involved in protein synthesis.

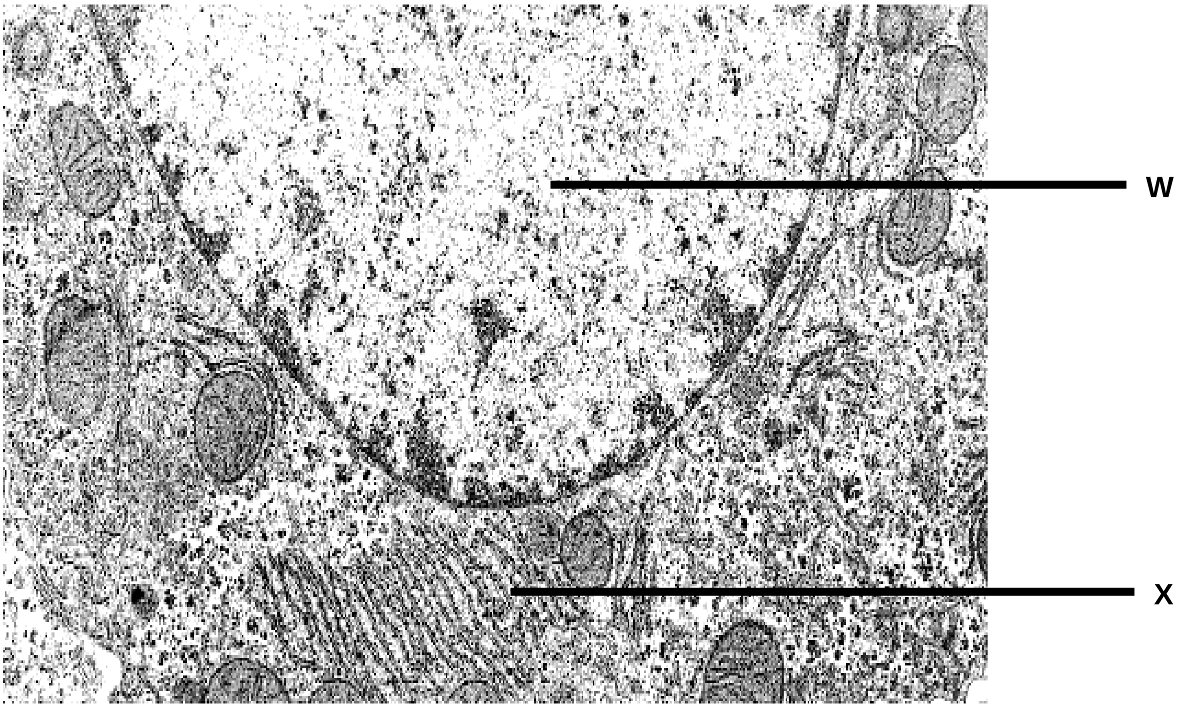

Fig. 1.1

(a) State three ways in which Organelle **W** differs from that of Organelle **X**. [3]

1. Organelle W is double membrane bound organelle whereas organelle X is a single membrane bound organelle.

2. There is absence of ribosomes studded on the membrane surface of Organelle W while there is presence of ribosomes studded on the membrane surface of Organelle X.

3. Organelle W contains the genetic material of the eukaryotic cell in the form of DNA whereas Organelle X translate the genetic material to synthesize protein.

An experiment was carried out to determine what happens to amino acids after they are absorbed by animal cells. The cells were incubated for 5 minutes in a medium containing radioactively labelled amino acids. The radioactive amino acids were then washed off and the cells were incubated in a medium containing only non-radioactive amino acids.

Samples of the cells were removed from the medium every five minutes for 40 minutes. For each sample, the levels of radioactivity in three different organelles, **P**, **Q** and **R**, were determined. The results of the experiment are shown in Fig. 1.2 below.

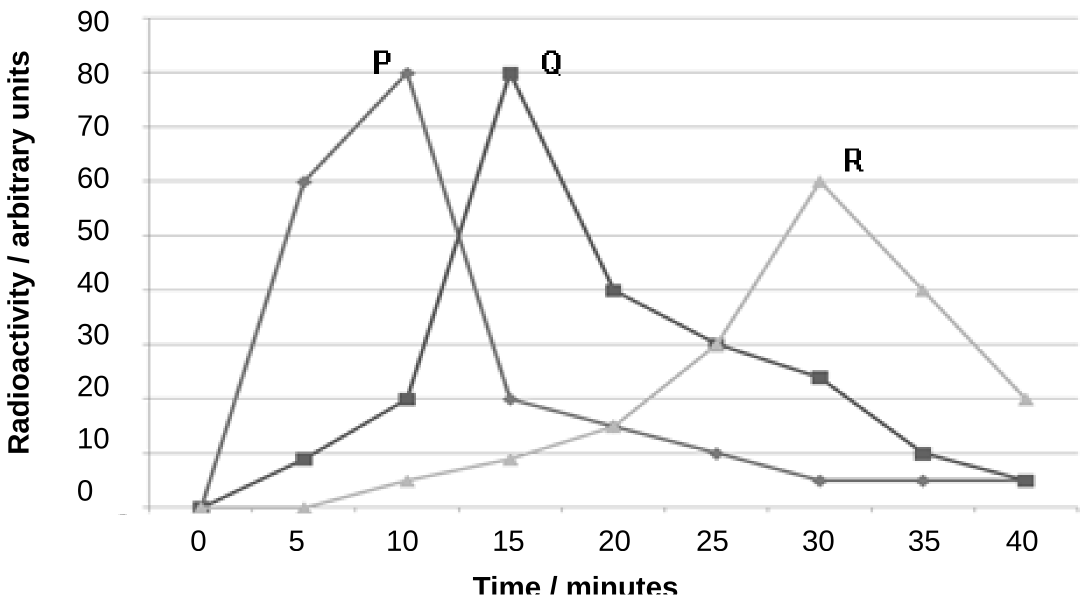

Fig. 1.2

(b) With reference to Fig. 1.2, explain the changes in the level of radioactivity of the three organelles with time. [3]

1. There is a rapid increase in radioactivity from 0 to 80 units (or of 8 arbitrary units / minute) detected in the first 10 minutes in P (rough endoplasmic reticulum) which indicated that the radioactive amino acids are transported to ribosomes on rough endoplasmic reticulum which are the sites of **protein synthesis** to form polypeptides.

2. Thereafter, radioactive proteins are transported to Q (Golgi apparatus) for **chemical modification and sorting** as shown by increase in radioactivity in Q from 20 to 80 units (or of 12 arbitrary units / minute) between 10-15 minutes with a corresponding decrease in radioactivity in P.

3. Proteins are then packaged into R (secretory vesicles) that bud off from trans face of Golgi apparatus as shown by increase in radioactivity in R from 10 to 60 units (or of 4 arbitrary units / minute) between 15 – 30 minutes with most proteins have been **secreted out of the cell via exocytosis** at 40 minutes followed by a decrease in radioactive in Q.

Fig. 1.3 shows the central dogma of molecular biology.

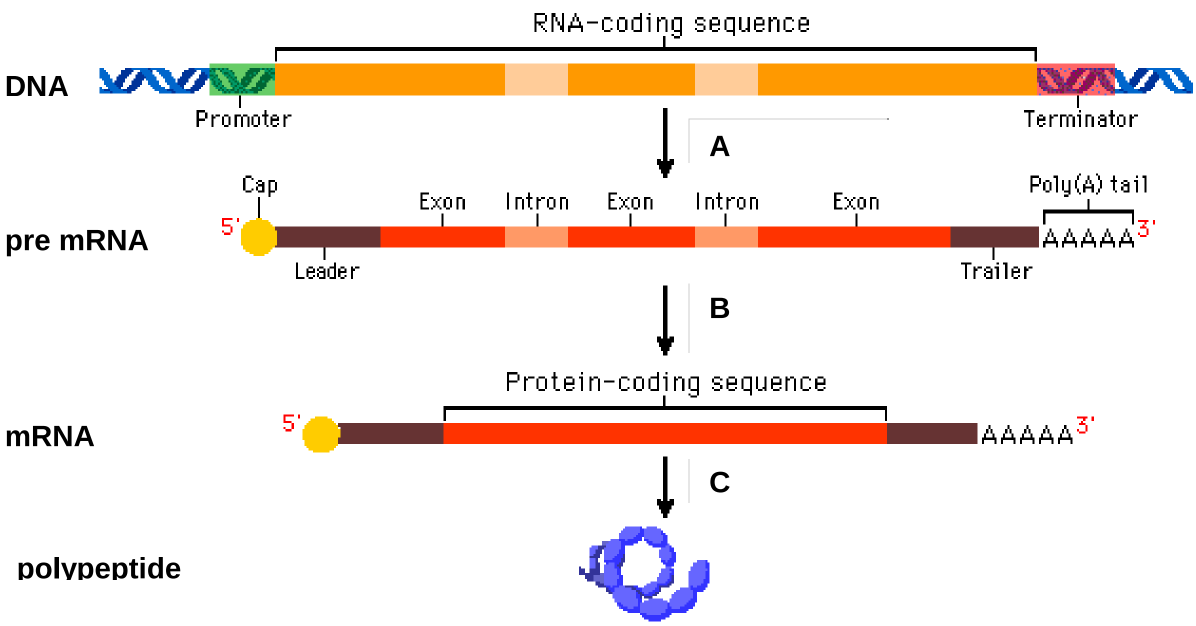

Fig 1.3

(c) **(i)** With reference to Fig 1.3, explain how the pre mRNA is processed to form mature mRNA for translation. [3]

1. The introns / non-coding sequences from the pre mRNA are excised.
2. The exons / coding sequences are spliced together.
3. This is catalysed by an enzyme complex called the splicesosome.

(ii) Transfer RNA (tRNA) is one of the molecules involved in translation. Describe one structural feature which adapts tRNA to its role in translation. [1]

1. There is a **binding site at 3' OH end / acceptor stem / with the sequence CCA** for a specific amino acid attachment corresponding to anticodon.
2. The **anticodon** on tRNA complementary base pair with codon of mRNA to translate the information from nucleotide sequence to amino acid sequence.
3. The **shape / 3D structure** of tRNA complementary to active site of aminoacyl tRNA synthetase for amino acid activation.
4. The shape / 3D structure of tRNA complementary to E, P and A sites of ribosomes for elongation of the amino acid sequence.
5. AVP;;

Any **1** ; structural feature must match the role for full credit.

[Total: 10]

2 In some species of the locusts, the female has a diploid chromosome number of 16 while the male locust has a diploid number of 15. The difference in chromosome number between the sexes is the result of a difference in the number of sex chromosomes (X chromosome). Female locusts have two X chromosomes (XX) and male locusts have only one X chromosome and no other sex chromosome (XO).

(a) Explain what is meant by the *"diploid chromosome number"*. [1]

1. It meant having two sets of chromosomes in a cell with one set from maternal and the other set from paternal.

(b) Explain why there is a need for reduction division prior to fertilisation in sexual reproduction. [2]

1. This is to decrease the ploidy level of the gametes to **half of that of the parental ploidy level**.

2. Therefore, this will ensure that the ploidy level of the zygote is **restored to back to parental ploidy level during fertilisation**.

(c) At the end of a meiotic division, state how many chromosomes would you expect in the gametes of:

(i) a male locust [1]

1. 7 chromosomes if gamete does not contain X chromosome
2. 8 chromosomes if gamete contain X-chromosome

(ii) a female locust [1]

8 chromosomes

(d) Darwinism is a theory of evolution based upon inherited variations in organisms and natural selection of fitter variants to produce species adapted to their habitats.

Twentieth-century biology added a theory of inheritance, the science of genetics, to give Neo-Darwinism. In the past 20 years the techniques of genetics and molecular biology have converged to provide both a remarkably detailed understanding of the genes that define the molecular composition of any organism and the ability to transfer genes from one species to another.

(i) With reference to the information above, explain how molecular evidence derived from techniques of genetics and molecular biology supports Darwin's theory of natural selection. [3]

1. There is **genetic variation** in the types of alleles and genes exist in the population due to mutation.

2. **Selection pressure** for the given environment selects for alleles/genes that give rise to **favourable traits** OR individuals with the favourable alleles/genes will be able to **survive till sexual maturity** and pass down their alleles to the next generation.

3. This will result in the **frequency of the genes and alleles change over time**.

(ii) One important aspect of modern theory of evolution is the presence of genetic variation in natural population. [2]

Describe the role of meiosis in natural selection.

1. Meiosis give rise to variation in the population generating **different types of gametes**.

2. With **different combination of alleles** from the same parental cells allows for selection of favourable traits for the given selection pressure.

[Total: 10]

3 In cats, the gene controlling cat colour is sex-linked. This gene has two alleles. In a particular colony of cats the coats of the males were either black or ginger and those of the females black, ginger or tortoiseshell (a patchwork black, ginger and cream).

Fig 3.1 shows a picture of a tortoiseshell cat.

Fig. 3.1

(a) With reference to the information above, state the genotypes according to their corresponding phenotypes of the different coats of the male cats using appropriate symbols. [1]

X^BY = black male

X^GY = ginger male

(b) Using a genetic diagram, determine the phenotypes expected in the progeny of a cross between a black female and a ginger male. [4]

Parental phenotypes : **black female x ginger male**

Parental genotypes : X^BX^B X^GY

Gametes X^B X^G Y

Punnett Square

	X^B	X^B
X^G	$X^B X^G$ Ginger female	$X^B X^G$ Ginger female
Y	X^B Y **Black male**	X^B Y **Black male**

Offspring genotypes : **1 X^BX^G : 1 X^BY**

Offspring phenotypes : **1 tortoishell female : 1 black male**

- 1 mark for correct genotype of parents.
- 1 mark for circling gamete (including those in punnet square)
- 1 mark for correct genotypes of all offspring
- 1 mark for associating the correct phenotype to genotype

(c) In a litter of kittens, half the number of females was tortoiseshell and the other half was black; half the number of males was black and the other half was ginger.

(i) State the colours of the parental male and female cats. [2]

Male **black** (X^BY)

Female **tortoiseshell** (X^BX^G)

(ii) Explain your answer in **(c)(i)**. [1]

1. Female parent contributed one of the **X chromosomes** to the male offspring, their mother must be **tortoiseshell** to have black and ginger male offspring
2. **black female offspring** indicates that male parent contributed **X^B allele thus male parent is black** (X^BY)

(e) A research team has also identified that external factors can also play a role in influencing the fur coat colour of the cats. [2]

Describe how a named external factor can influence the fur coat colours of cats.

1. A named factor: UV radiation, diet, AVP;;
2. High exposure to UV radiation may have result in darkening of the fur coat due to increase pigments production.

[Total: 10]

4 Restriction enzymes cut DNA molecules only at specific sites with particular base sequences. The target sites for the enzyme **A**, **B** and **C** are shown in Table 4.1 below. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

Name of restriction enzyme	DNA base sequence of target site
A	-C-C-C G-G-G- -G-G-G C-C-C-
B	-C- C-G-G- -G-G-C- C-
C	-A -A-G-G-T-T- -T-T-C-G-A- A-

Table 4.1

The target base sequences for all three restriction enzymes in Table 1.1 are palindromic.

(a) **(i)** With reference to Table 1.1, explain what is meant by the term palindromic. [1]

1. The DNA sequence on complementary strands read the same in opposite directions **OR**

2. When reading the top DNA sequence from left to right is the same as reading the bottom DNA sequence from right to left.

(ii) Suggest why it is preferable that restriction enzyme target sequences be palindromic. [1]

This is to allow restriction enzymes to cut both strands at the same time to produce fragments.

Reject: produce sticky ends as not all restriction enzymes such as restriction enzyme A produce sticky end.

Restriction digests were performed separately on pure samples of a plasmid using 4 different restriction enzymes, RE1, RE2, RE3 and RE4.

Lanes 1, 2, 3 and 4 of Fig 4.2 show the result of the restriction digests using restriction enzymes, RE1, RE2, RE3 and RE4 respectively. A DNA ladder (with known fragment lengths) was loaded in Lane 5.

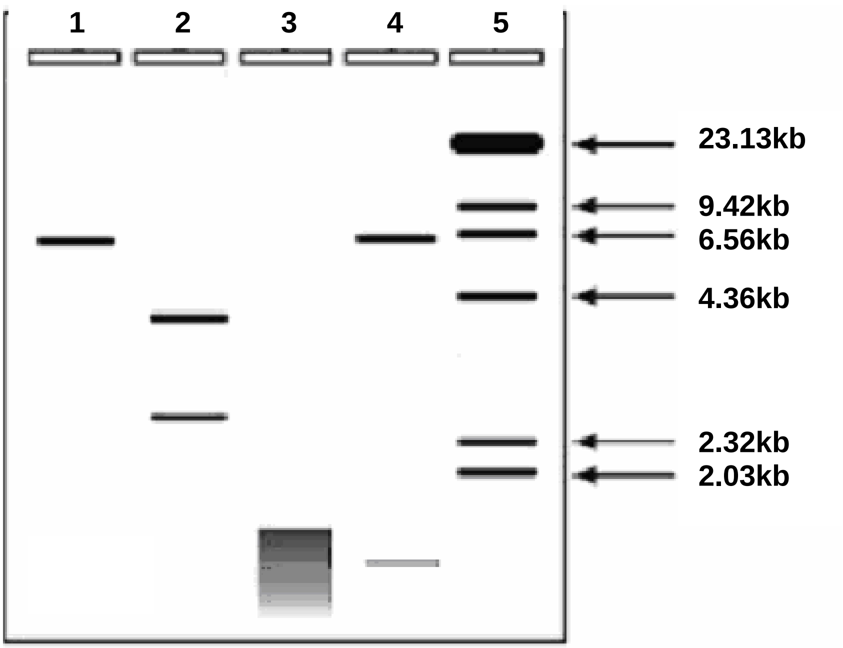

Fig 4.2

(b) **(i)** Using Fig 4.2, identify the approximate size of the plasmid use in this genetic engineering. [1]

Accept 6 to 6.56 kb

Reject: anything longer than 6.56kb as shown in lane 1 of the gel electrophoresis

(ii) State one limitation in using a plasmid vector. [1]

There is a limit to the size of the DNA insert that it can carry / insert during genetic engineering.

(iii) If this experiment was performed without any error, suggest a reason why a smear was obtained in lane 3. [2]

1. RE3 used must have cut the plasmid many times resulting in many smaller fragments OR plasmids contain many RE3 restriction sites resulting in multiple small fragments.

2. These fragments are similar in size that they cannot be distinguished between adjacent bands / that they have the same migration rate and distance.

(iv) Polymerase chain reaction (PCR) was performed to amplify the quantity of the plasmid used in restriction digest. [2]

Describe the role of the two forward and reverse primers in PCR.

1. The forward and reverse primers are complementary to the 3' flanking sequences of the region to be amplified.

2. The primers anneal to the 3'flanking sequences by complementary base pairing through the formation of hydrogen bonds.

(c) The plasmid was found to only contain one selectable marker. [2]

Suggest why a plasmid containing one selectable marker is not ideal for use in cloning when we want to insert a foreign gene into *E.coli*.

1. One selectable marker is to allow for selection for transformed bacterial cells that have taken up the plasmid.

2. The other selectable marker is to identify for transgenic bacteria containing gene of interest via insertional inactivation.

[Total: 10]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where applicable.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

5 **(a)** Glyceraldehyde 3-phosphate is the precursor to forming glucose during Calvin Cycle. Glucose is polymerised into starch, a macromolecule for storage. [6]

Outline how glyceraldehyde 3-phosphate is made in the Calvin cycle and converted into starch for storage.

How glyceraldehyde 3-phosphate is made in the Calvin cycle

1. Requires one ATP and one NADPH per G3P formed.

2. Each CO2 is attached/**fixed** to ribulose bisphosphate (RuBP) to form 2 molecule of glycerate-3-phosphate (GP) catalysed by ribulose bisphosphate carboxylase/Rubisco.

3. GP phosphorylated to form glycerate bisphosphate /1, 3-bisphosphoglycerate (BPG) by phosphorylase.

4. Carboxyl group reduced to aldehyde group by a pair of electrons, forming glyceraldehyde-3-phosphate (G3P)

(Max 4)

How G3P converted into starch

1. Two molecules of G3P formed one molecule of α-glucose

2. Glucose joined by α(1, 4) and α (1, 6) glycosidic bond during condensation reaction to formed starch (with amylose and amylopectin)

(Max 2)

(b) There are many factors that can affect the rate of photosynthesis in plants. [8]

With reference to the term ‘limiting factor’, explain how two named factors can have effects on photosynthesis.

A limiting factor is a factor, which is in the shortest supply, that **controls/determines how fast or slow the rate** of photosynthesis process is. [2]

Rate of photosynthesis can limited by [Any named two factors]:

- Light intensity [max 3]
  1. **Affect the light-dependent stage of photosynthesis** as low light intensity, light intensity is the limiting factor;
  2. Rate of photosynthesis increases proportionally with increasing light intensity; [**correct trend**]
  3. At high intensities, rate of photosynthesis decreases as further increase in light intensity results in light saturation point being reached and **beyond further increase in light intensity will have no effect on the rate of photosynthesis** as light may bleach chlorophyll and retard photosynthesis
- Carbon dioxide [max 3]
  1. Major limiting factor as atmospheric CO_2 concentration is low as **CO_2 is needed in the light-independent stage / Calvin cycle**;
  2. Rate of photosynthesis increases proportionally with an increase in CO_2 concentration; [**correct trend**]
  3. It is fixed_by combining with_ribulose-1,5-bisphosphate (RuBP) by the enzyme ribulose bisphosphate carboxylase oxygenase (RUBISCO) and reduced to carbohydrate / sugar
- Temperature [max 3]
  1. Affect mainly the **light-independent stage/ Calvin cycle** as it is enzyme-controlled;
  2. Rate of photosynthesis increases with an increase in temperature where rate doubles for each rise of 10^{o}C until optimum temperature is reached; [**correct trend**]
  3. At higher temperatures **beyond optimum, rate of photosynthesis decreases** as further increase in temperature as enzymes involved in the light-independent / dark reaction stage start to **denature**.
- Oxygen [max 3]
  1. Affect mostly the **light-independent stage / Calvin cycle**;
  2. Rate of photosynthesis decreases with increasing oxygen concentration; **[correct trend]**
  3. **Oxygen competes with CO_2** for the active site of the enzyme ribulose bisphosphate carboxylase oxygenase (RUBISCO).

(c) It might be possible to produce genetically modified plants with increased photosynthetic enzymes with enhanced efficiency. [6]

Discuss the ethical implications of genetically modified plants.

1. GM plants grown as crops may lead to consumers **having allergies as foreign proteins are produced in the plants**, companies need to label their GM crops for consumers to make informed choices;

2. Animal genes may be introduced to plant genomes, leading to **concern of vegetarians and some religious groups** which followers are not allowed to consume certain animals;

3. Genetic modifications of certain plants may **affect food chains / certain animals that feed on the plants**. E.g. of genetic modifications of plant which affect food chain: Bt corn plant pollen blown onto milkweed plants can be ingested by the Monarch Butterfly;

4. **Cross-pollination of GM plants with other non-crop plants lead to transfer of genes**, affecting ecological balance; E.g of transfer of genes to non-crop plants leading to ecological balance disrupted: Bt gene transferred to non-crop plants leading to these plants producing toxin killing desirable insects such as Monarch butterfly;

5. GM crops lead to **benefits that rich countries can enjoy** due to more financial resources at the expense of poorer countries;

6. GM crops can produce higher quality food to allow **large companies that develop the technology to out-compete small scale farmers**; Example: Golden rice, Bt corn;

7. **Tampering with nature**, where the mixing of genes among species may be seen as violation of organisms natural intrinsic values, crossing species boundaries;

8. AVP;

(max 6)

[Total: 20]

Or

6 **(a)** The linked reaction is an important process between glycolysis and Krebs cycle in aerobic respiration. [6]

Outline the linked reaction and state where it occurs.

1. One molecule of **CO_2 is removed** from pyruvate (3C) – decarboxylation.
2. Oxidation by **dehydrogenation occurs to the remaining 2C fragment**, yielding NADH.
3. **Coenzyme A (CoA) is added** to the remaining 2C acetyl group to form acetyl coenzyme A / acetyl CoA (2C).
4. The process of converting pyruvate to acetyl CoA, whereby it loses a CO_2 and is also oxidised, is termed as **oxidative decarboxylation**.
5. This process occurs in the **mitochondrial matrix** and is catalysed by **pyruvate dehydrogenase**.
6. Acetyl-CoA moves into the Krebs cycle.

(b) Some plants will die from alcohol accumulation when submerged in water for prolonged period of time. Athletes may accumulate lactic acid in their muscles when they engaged in strenuous activity. [8]

Explain how plants and humans accumulate alcohol and lactic acid respectively.

1. In the absence of oxygen, glycolysis can occur but oxidative phosphorylation is blocked because O_2 is the final electron acceptor in oxidative phosphorylation.
2. Therefore **NAD^+ is not regenerated** from oxidation of NADH hence conversion of pyruvate to acetyl CoA, and Krebs cycle are blocked.
3. In anaerobic condition, **glycolysis occurs** and glucose is broken down to pyruvate and NADH and glycolysis, too, would become blocked unless NAD^+ becomes available.
4. To regenerate NAD^+, NADH is oxidised by pyruvate catalysed by pyruvate dehydrogenase to give **lactate** (in process called **lactate fermentation**) in human
5. However, in plant, NAD^+ is regenerated by reducing pyruvate to acetaldehyde and subsequently **alcohol** (in a process called **alcoholic fermentation**) catalysed by alcohol dehydrogenase.
6. Both alcoholic fermentation and lactate fermentation do **not** yield ATP molecules.
7. They are mere mechanisms to **regenerate NAD^+ from NADH** to keep glycolysis going.
8. Only a **small amount of NAD^+** is regenerated from alcoholic fermentation and lactate fermentation, and this amount is sufficient for glycolysis but not the conversion of pyruvate to acetyl CoA and the Krebs cycle.

(c) Oxidative phosphorylation produces a total of 28 ATP from re-oxidising NADH and $FADH_2$. [6]

Discuss the roles of NADH and $FADH_2$ in aerobic respiration.

1. NAD and $FADH_2$ are coenzymes that **removes high energy protons and electrons** from the Krebs cycle and **donates them to the electron transport** chain in the inner membrane of the mitochondrion so that oxidative phosphorylation can occur and produce ATP.

2. In glycolysis, link reaction and Krebs cycle, NAD is reduced to NADH

3. In Krebs cycle, FAD is reduced to $FADH_2$

4. In oxidative phosphrylation process, where NADH and $FADH_2$ are re-oxidised, releasing protons (H^+) and electrons which will pass through the ETC producing energy which is used to pump protons across the membrane to generate the proton gradient.

5. At the same time, **NAD and FAD is regenerated and fed back into glycolysis and Krebs cycle**.

6. The role of oxygen is to be the final electron acceptor from the oxidation of NADH and $FADH_2$ to produce water as by-product.

[Total: 20]